Sequence Diagram

- Purpose of Sequence diagram
- Types of objects
- Creating sequence diagram

Learning Outcomes

- Explain the purpose of sequence diagram
- Identify the objects necessary to create a sequence diagram
- Create sequence diagram for each use case

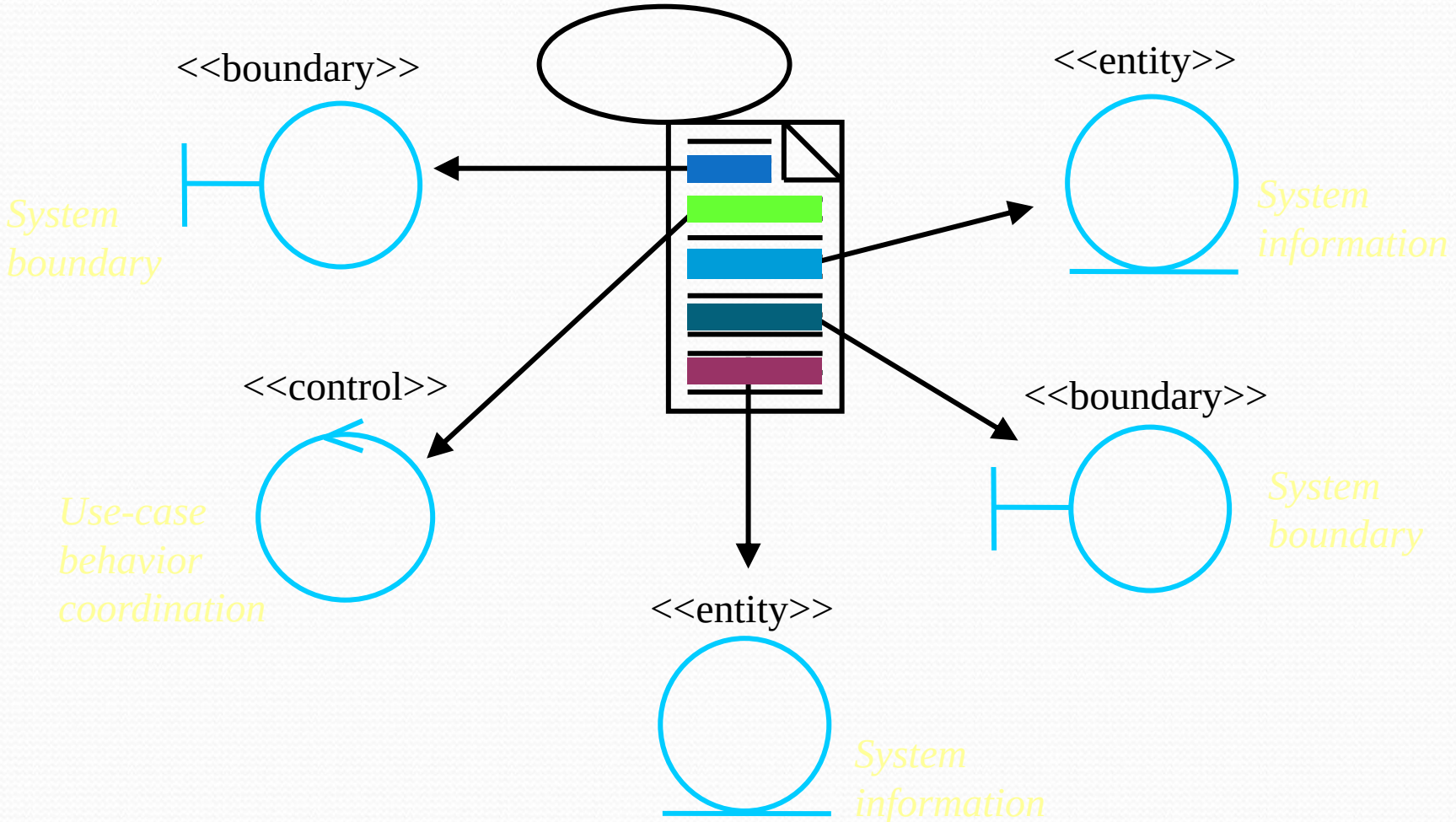
Sequence Diagrams

- Used to model the sequence of interactions among objects needed to realize a use case.
- Tie use cases with objects.
- Shows how the behavior of a use case is distributed among its participating objects.
- Allow developers to find missing objects in the system specification.

Sequence Diagrams…

- Object Types
  - Entity objects
  - Boundary objects
  - Control objects

What Is an Analysis Class?



Object Types

- Entity Objects
  - Represent the persistent information tracked by the system (Application domain objects, “Business objects”)
- Boundary Objects
  - Represent the interaction between the user and the system
- Control Objects:
  - Represent the control tasks performed by the system
- Having three types of objects leads to models that are more resilient to change.
  - The boundary of a system changes more likely than the control
  - The control of the system change more likely than the application domain
- Object types originated in Smalltalk:
  - Model, View, Controller (MVC)

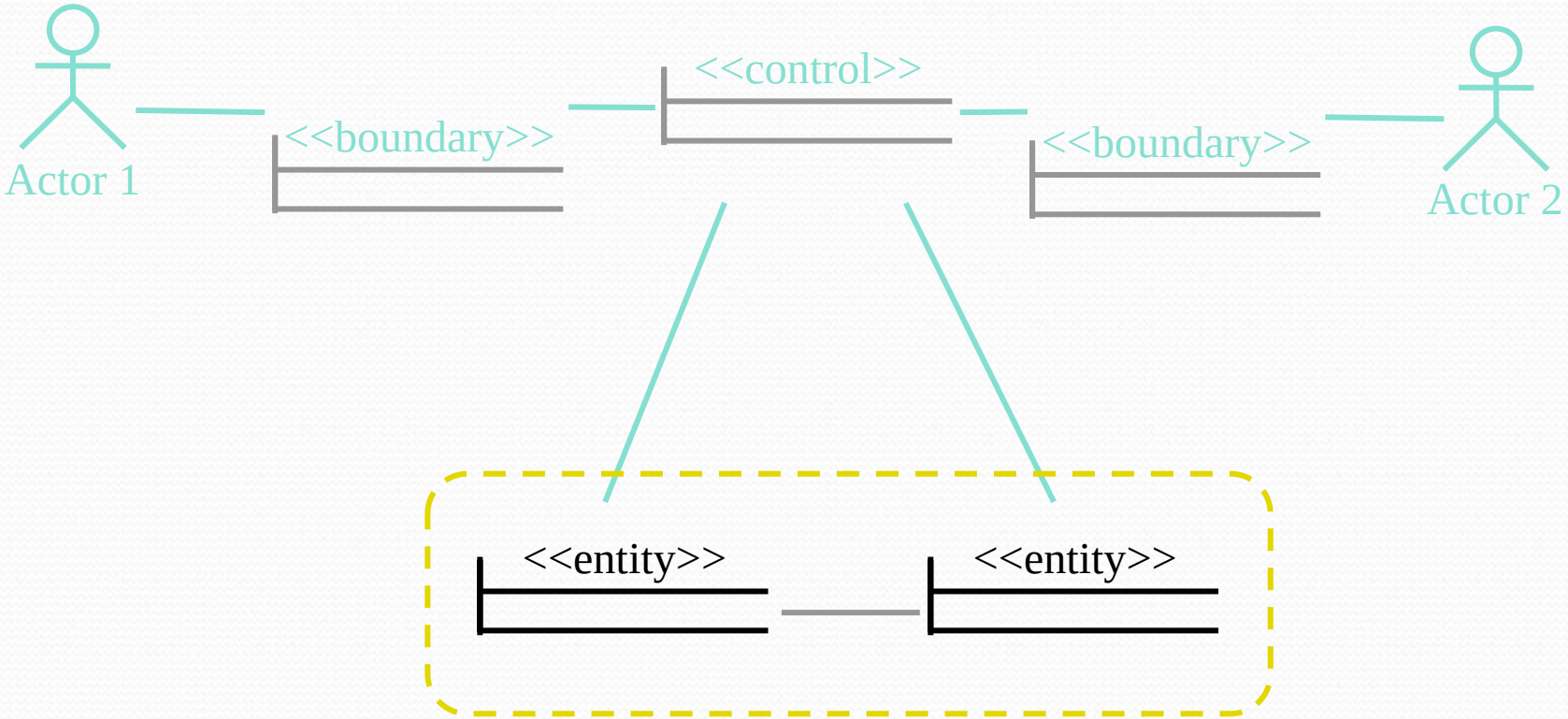
Entity Objects

- Represent the persistent information tracked by the system (Application domain objects, “Business objects”)
- Participating objects are found by examining each use case and identifying candidate objects.
- Natural language analysis provides an intuitive set of heuristics for identifying objects, attributes, and associations from a system specification.
- Abbot’s heuristics maps parts of speech (e.g., nouns, having verbs, being verbs, adjectives) to model components (e.g., objects, operations, inheritance, relationships, classes).

Entity Objects…

- Natural language has the advantage of :
  - Focusing on the user's terms
- Natural language has the disadvantage of :
  - Quality of the model depends highly on the style of writing of the analyst (e.g., consistency of terms used)
  - There are for example many more nouns than relevant classes. Many nouns correspond to attributes or synonyms for other nouns.

The Role of an Entity Class



Store and manage information in the system

Example: Finding Entity Classes

- Use use-case flow of events as input
- Traditional, filtering nouns approach
  - Underline noun clauses in the use-case flow of events
  - Remove redundant candidates
  - Remove vague candidates
  - Remove implementation constructs
  - Remove attributes (save for later)
  - Remove operations

Example: Candidate Entity Classes

- Register for Courses (Create registration)



CourseOffering



CoursesRegistered



Student

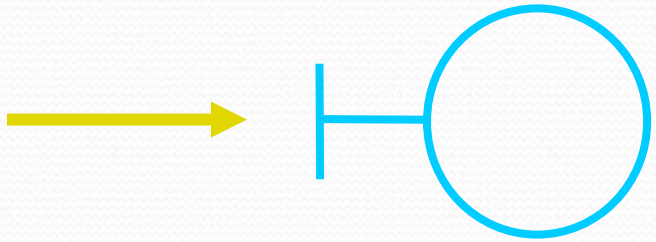
Boundary Objects

- Boundary objects represent the system interface with the actors.
- In each use case, each actor interacts with at least one boundary objects.
- The boundary object collects the information from the actor and translates it into an interface neutral form that can be used by the entity objects and also by the control objects.
- Boundary objects model the user interface at a coarse level

What Is a Boundary Class?

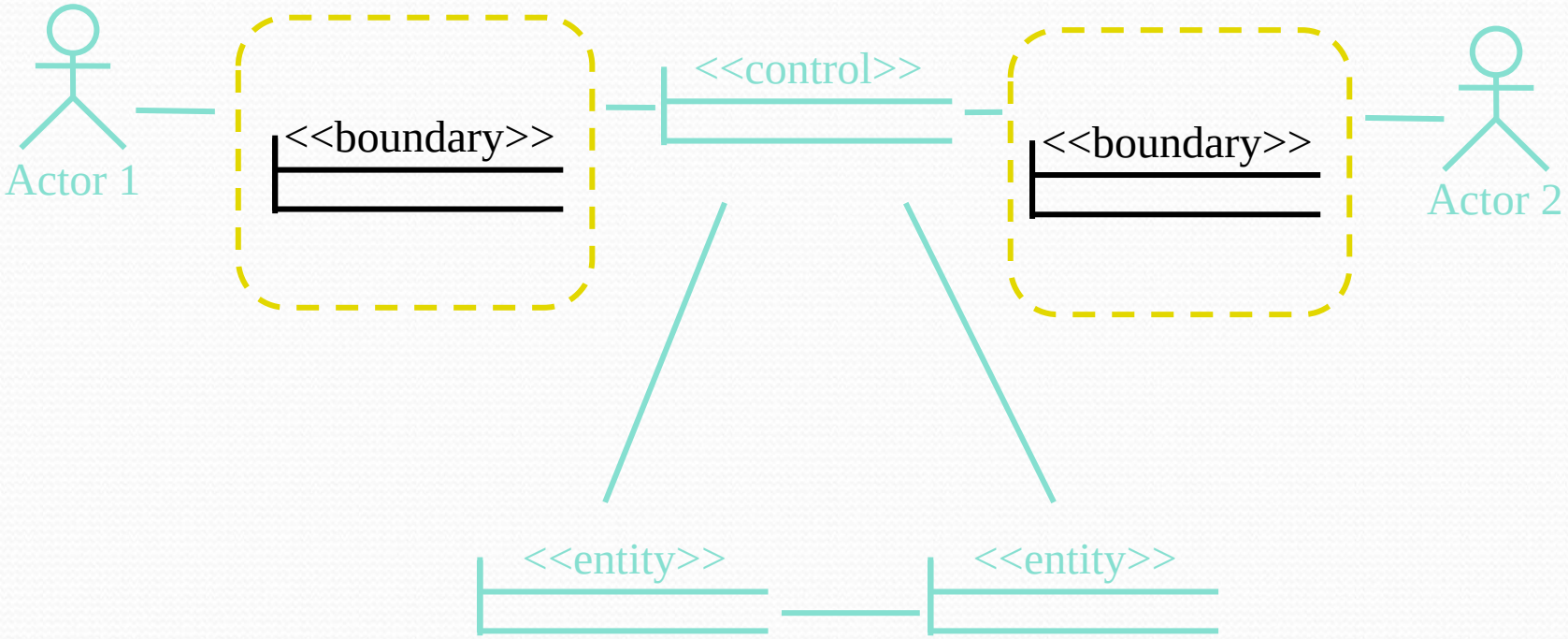
- Intermediates between the interface and something outside the system
- Several Types
  - User interface classes
  - System interface classes
  - Device interface classes
- *One boundary class per actor/use-case pair*

Analysis class stereotype



Environment Dependent

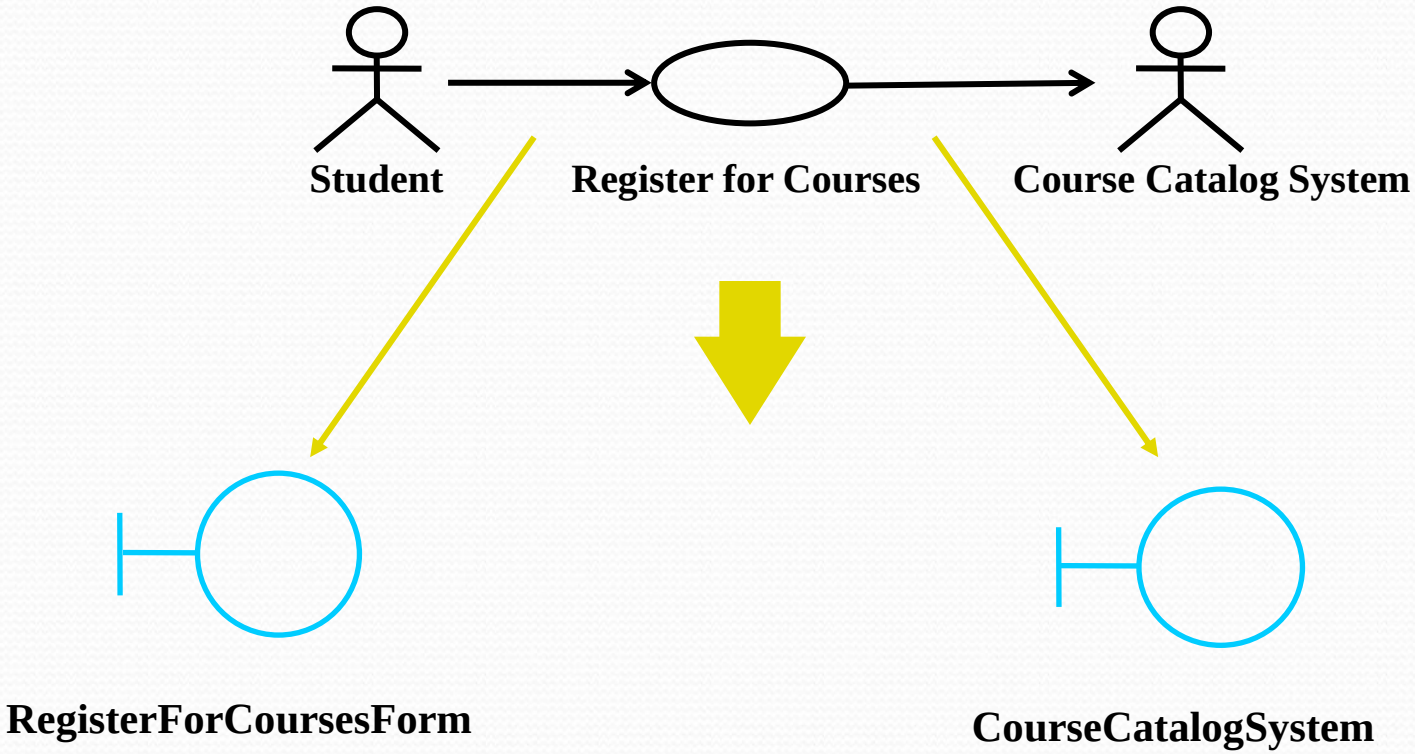
The Role of a Boundary Class



Model interaction between the system and its environment

Example: Finding Boundary Classes

- One boundary class per actor/use case pair



Guidelines: Boundary Class

- User Interface Classes
  - Concentrate on what information is presented to the user
  - Do NOT concentrate on the UI details
- System and Device Interface Classes
  - Concentrate on what protocols must be defined
  - Do NOT concentrate on how the protocols will be implemented

Concentrate on the responsibilities, not the details!

Boundary Objects…

- Heuristics for identifying boundary objects
  - Identify forms and windows the users need to enter data into the system. e.g., EmergenecyReportForm
  - Identify notices and messages the system uses to respond to the user (e.g., AcknowledgementNotice).
  - Do not model the visual aspects of the interface with boundary objects.
  - Use the user's terms for describing interfaces as opposed to terms from the implementation technology.

Boundary Objects…

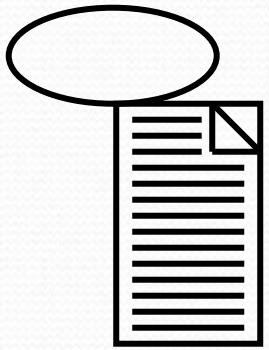
- Example:
  - Examining the ReportEmergency use case, we find the following boundary objects:
    - AcknowlegementNotice
    - EmergencyReportForm
    - IncidentForm
  - This provides the necessary interfaces between actors and the system.

Control Objects

- The next task is to describe the order in which the interactions between the actors and the system occur.
- Control objects are responsible for coordinating boundary and entity objects.
- A control object is usually created at the beginning of a use case and ceases to exist at its end.
- A control object is responsible for collecting information from the boundary objects and dispatching it to entity objects.

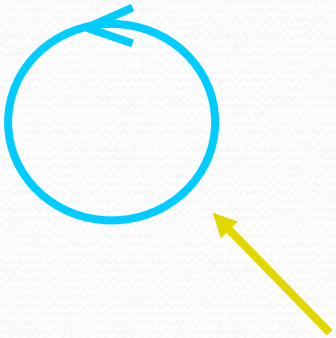
What Is a Control Class?

- Use-case behavior coordinator
  - More complex use cases generally require one or more control cases



Use Case

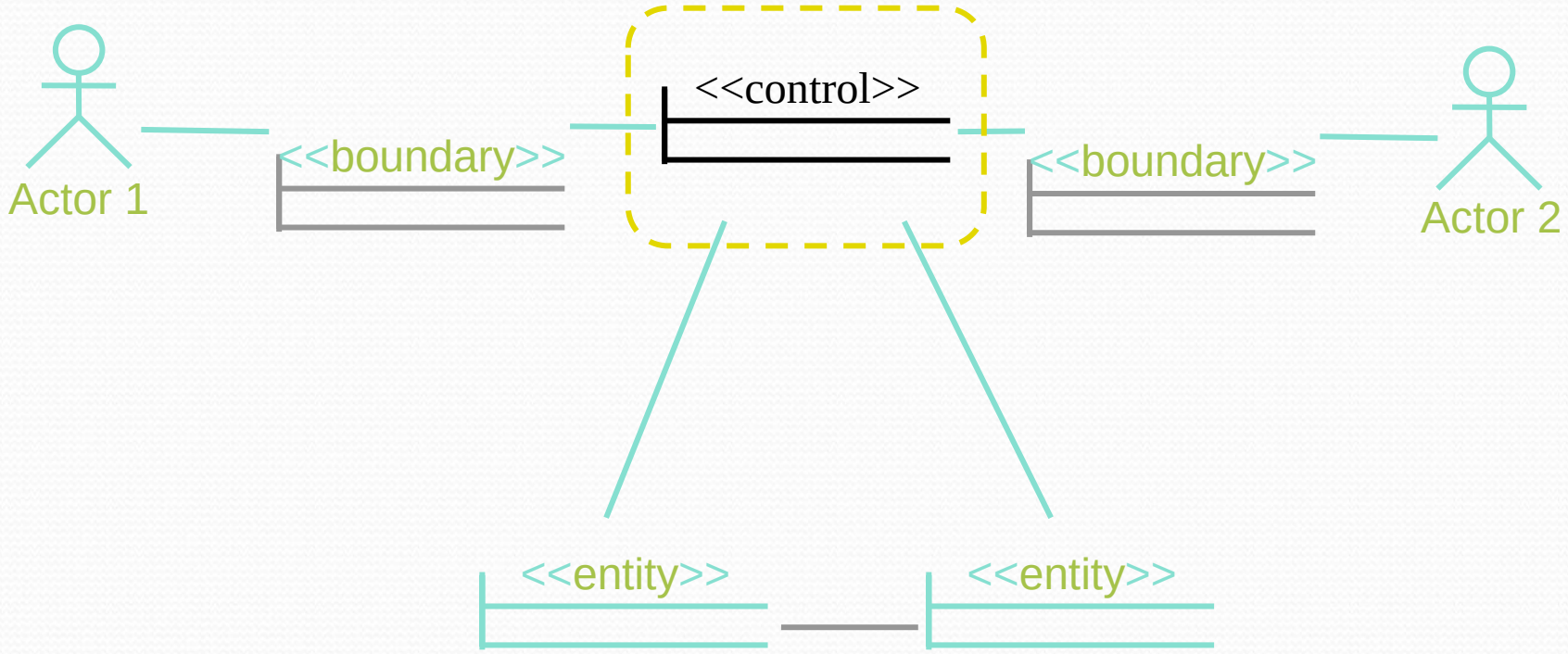




Analysis class stereotype

Use-case dependent, Environment independent

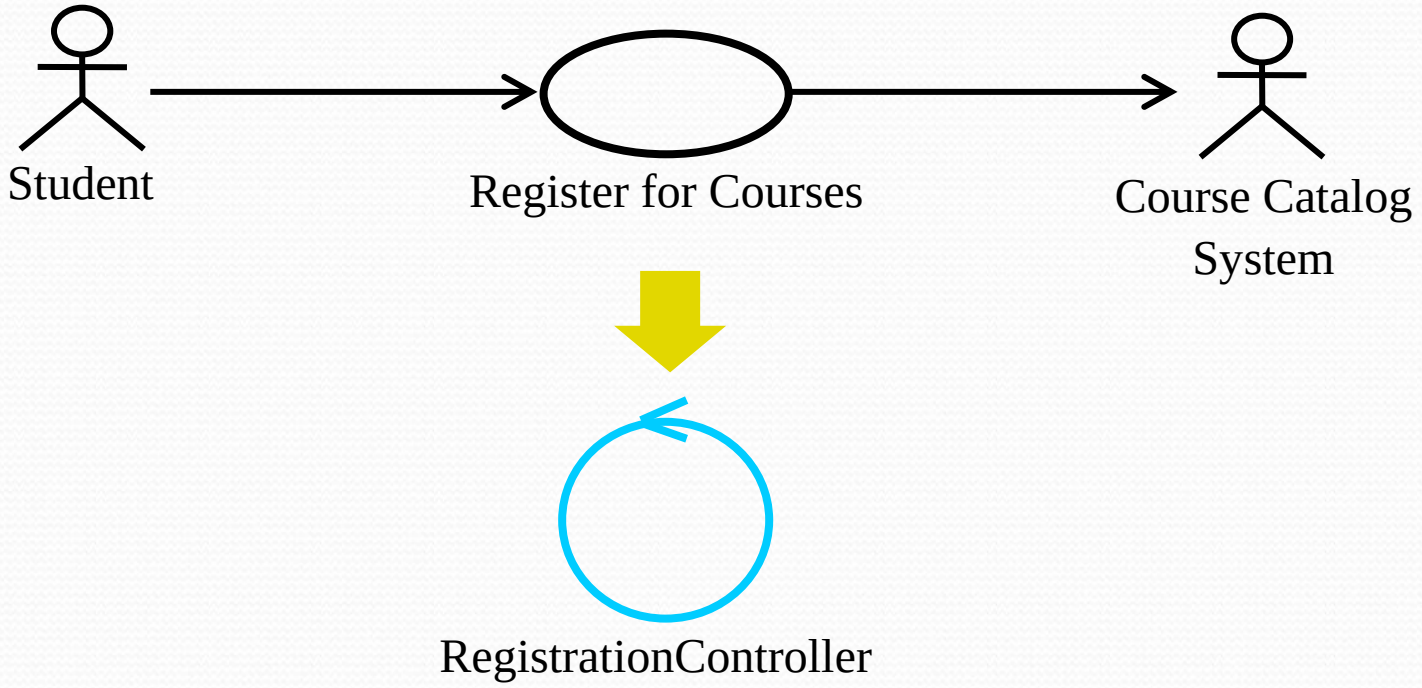
The Role of a Control Class



Coordinate the use-case behavior

Finding Control Classes

- In general, identify one control class per use case.
  - As analysis continues, a complex use case's control class may evolve into more than one class

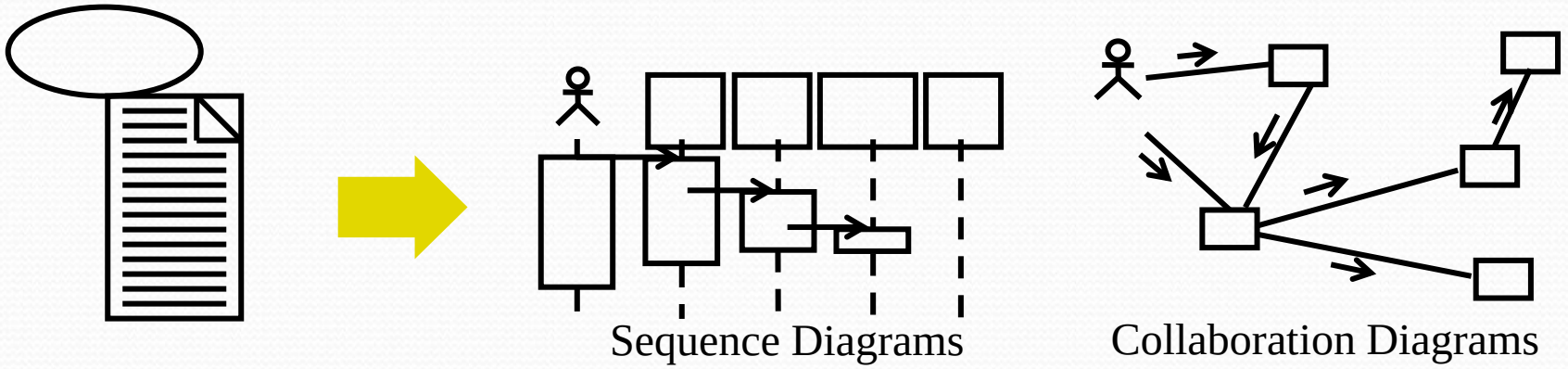


Control Objects…

- Heuristics for identifying control objects
  - Identify one or more control object per use case
  - Identify one control object per actor in the use case
  - The life span of a control object should be extent of the use case or the extent of a user session
- Example:
  - Examining the ReportEmergency use case, we find a control object for each actor.
    - ReportEmergnecyContrl for the field officer
    - ManageEmergencyControl for the dispatcher

Distribute Use-Case Behavior to Classes

- For each use-case flow of events:
  - Identify analysis classes
  - Allocate use-case responsibilities to analysis classes
  - Model analysis class interactions in Interaction diagrams



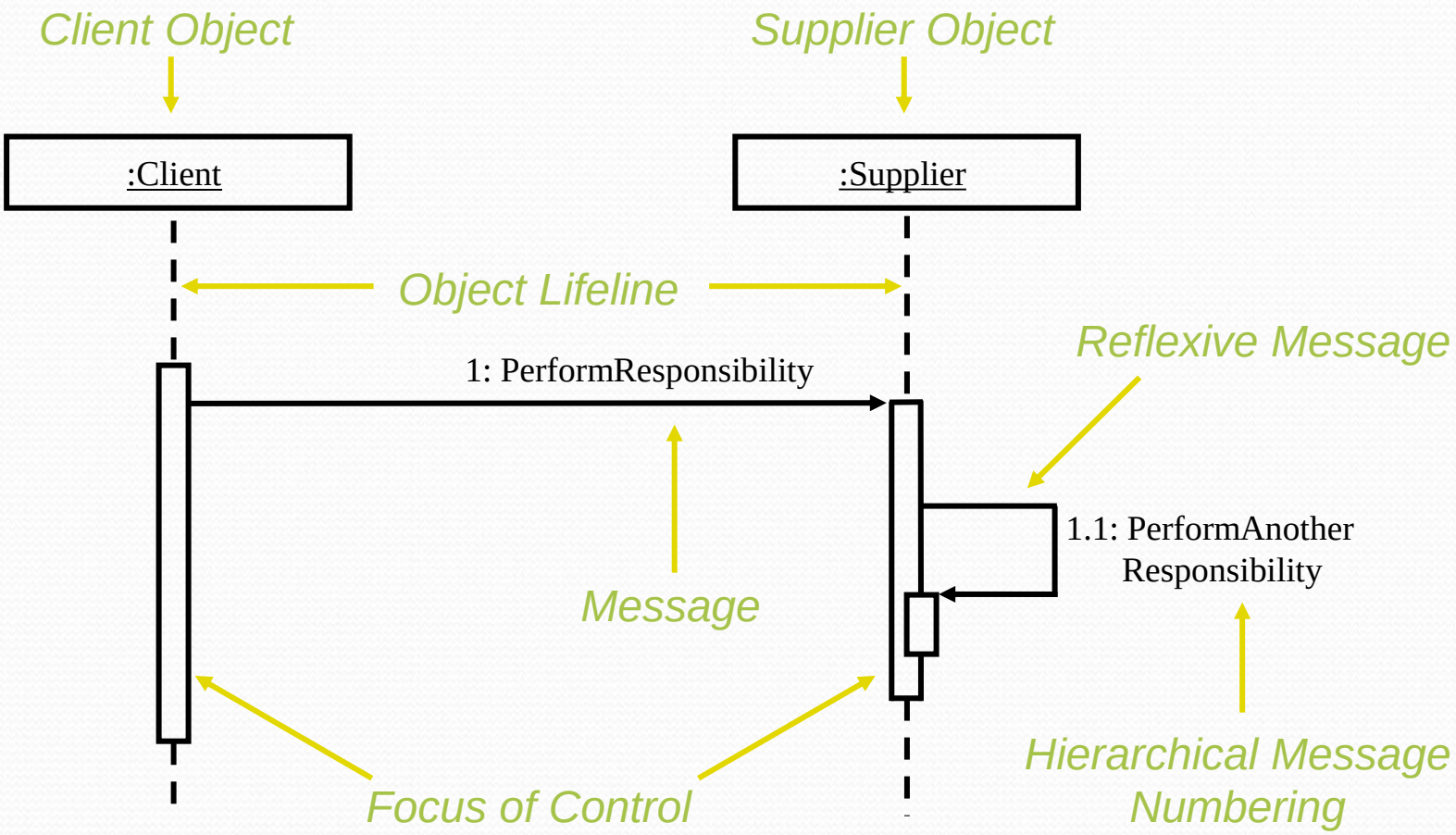
Use Case **Use-Case Realization**

Heuristics for Sequence Diagrams

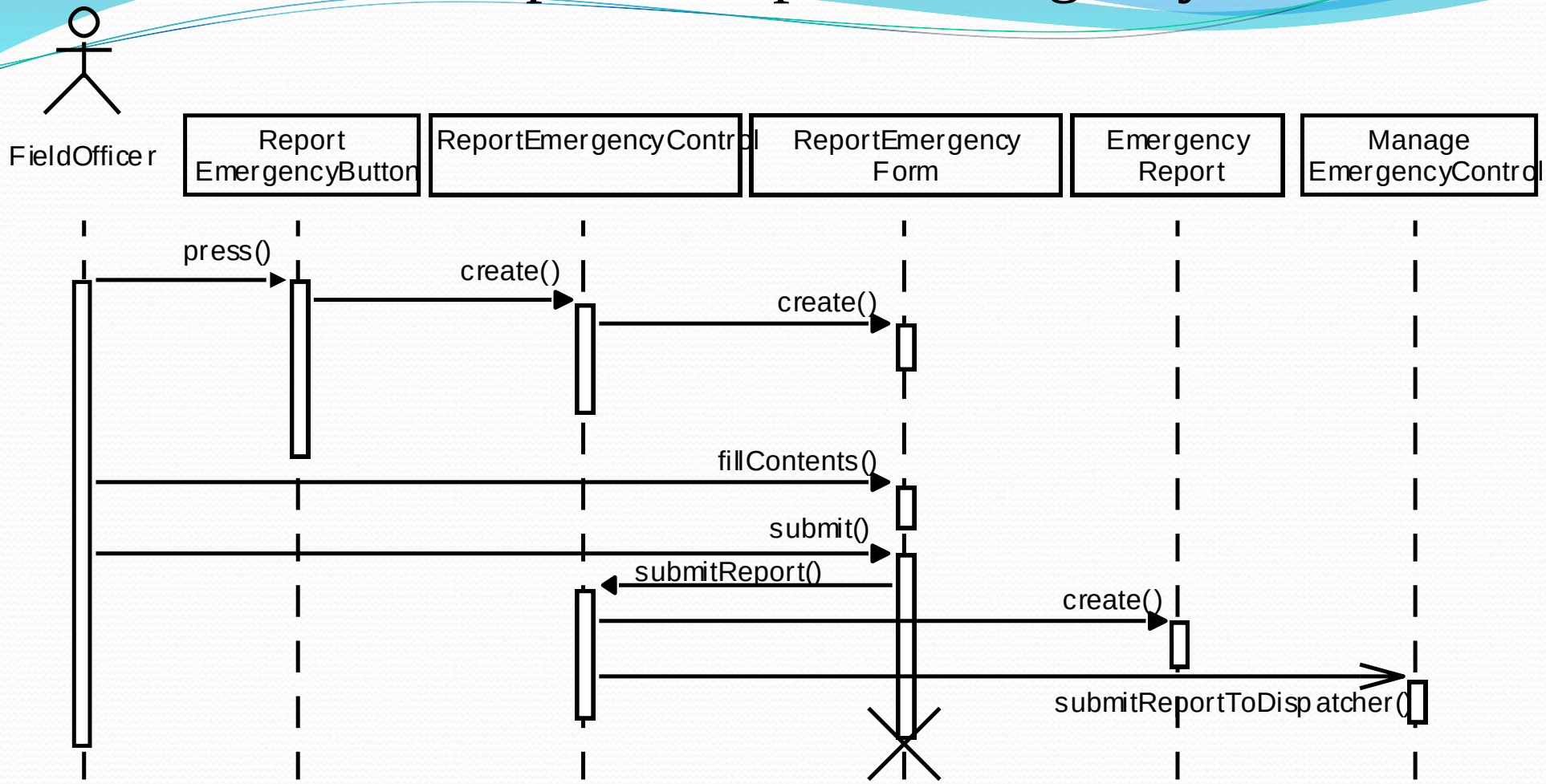
- Layout:
  - 1st column: Should correspond to the actor who initiated the use case
  - 2nd column: Should be a boundary object
  - 3rd column: Should be the control object that manages the rest of the use case
- Creation:
  - Control objects are created at the initiation of a use case
  - Boundary objects are created by control objects
- Access:
  - Entity objects are accessed by control and boundary objects,
  - Entity objects should never call boundary or control objects

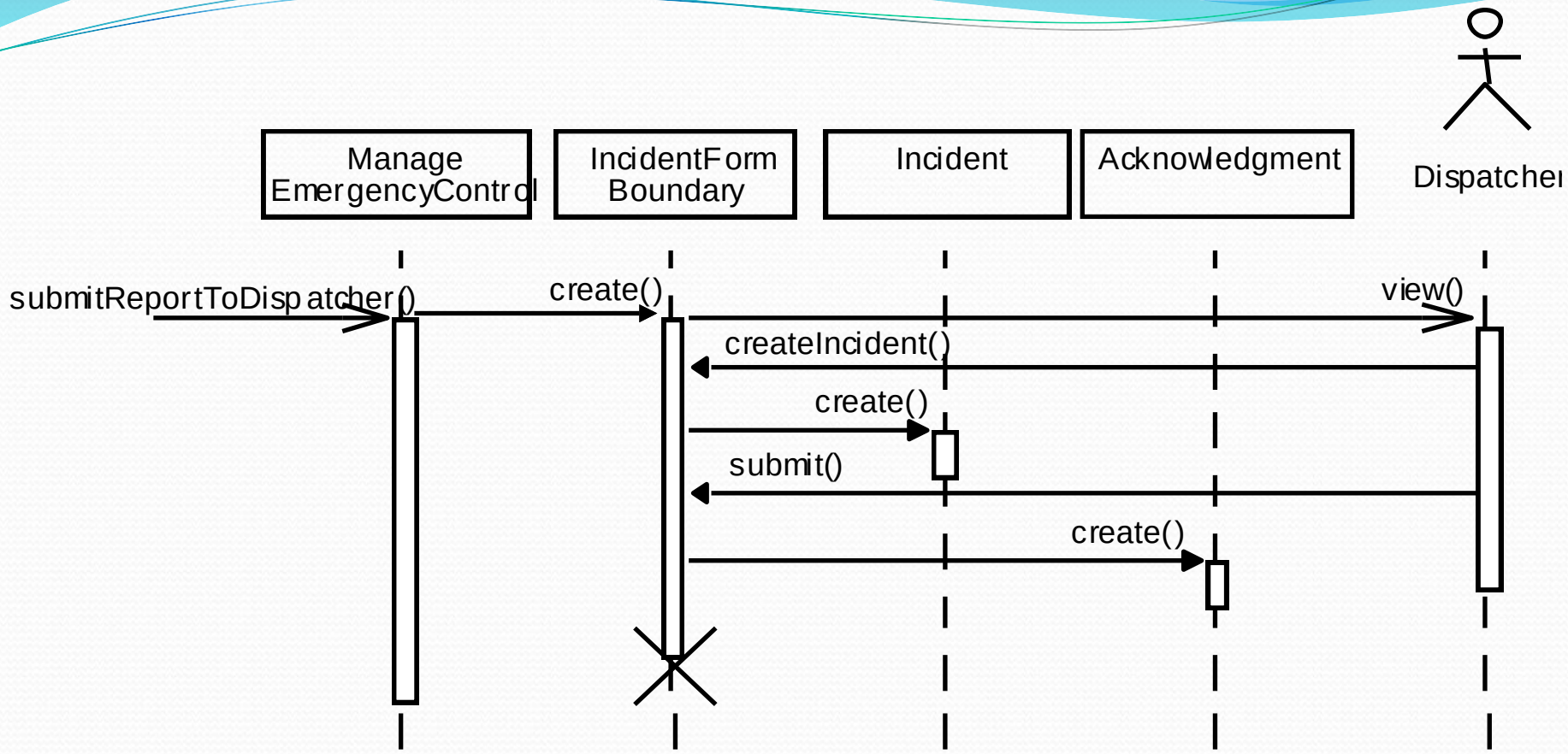
The Anatomy of Sequence Diagrams

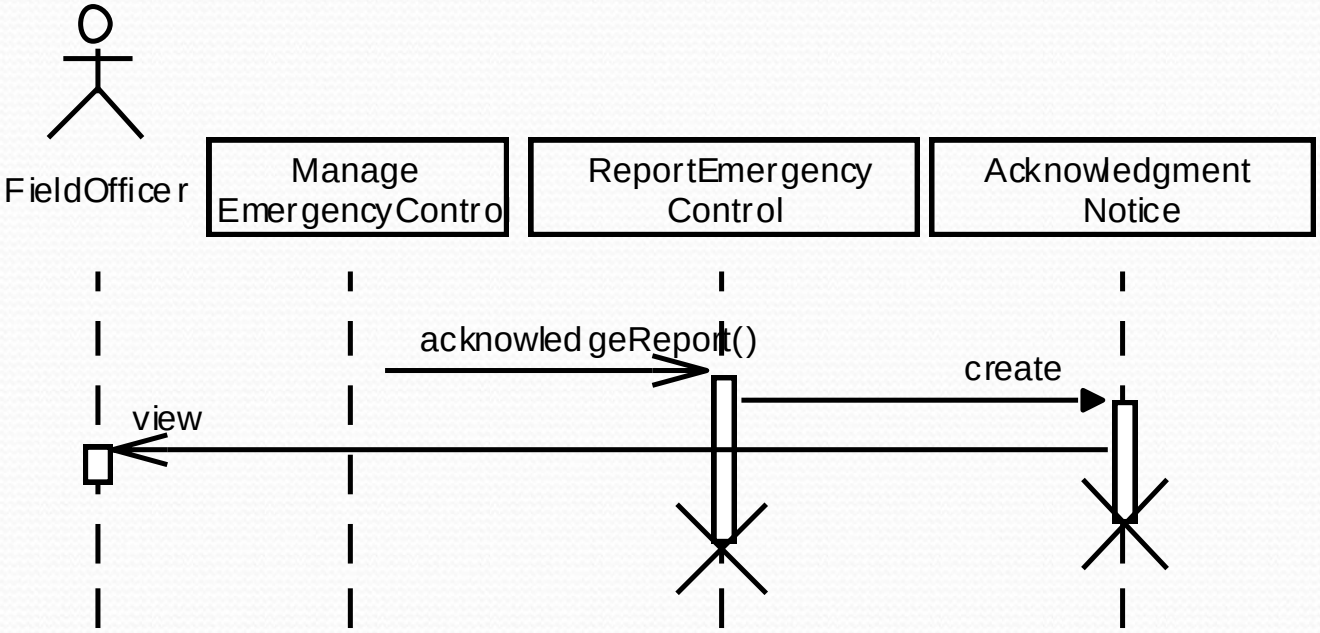
This is a sample script.



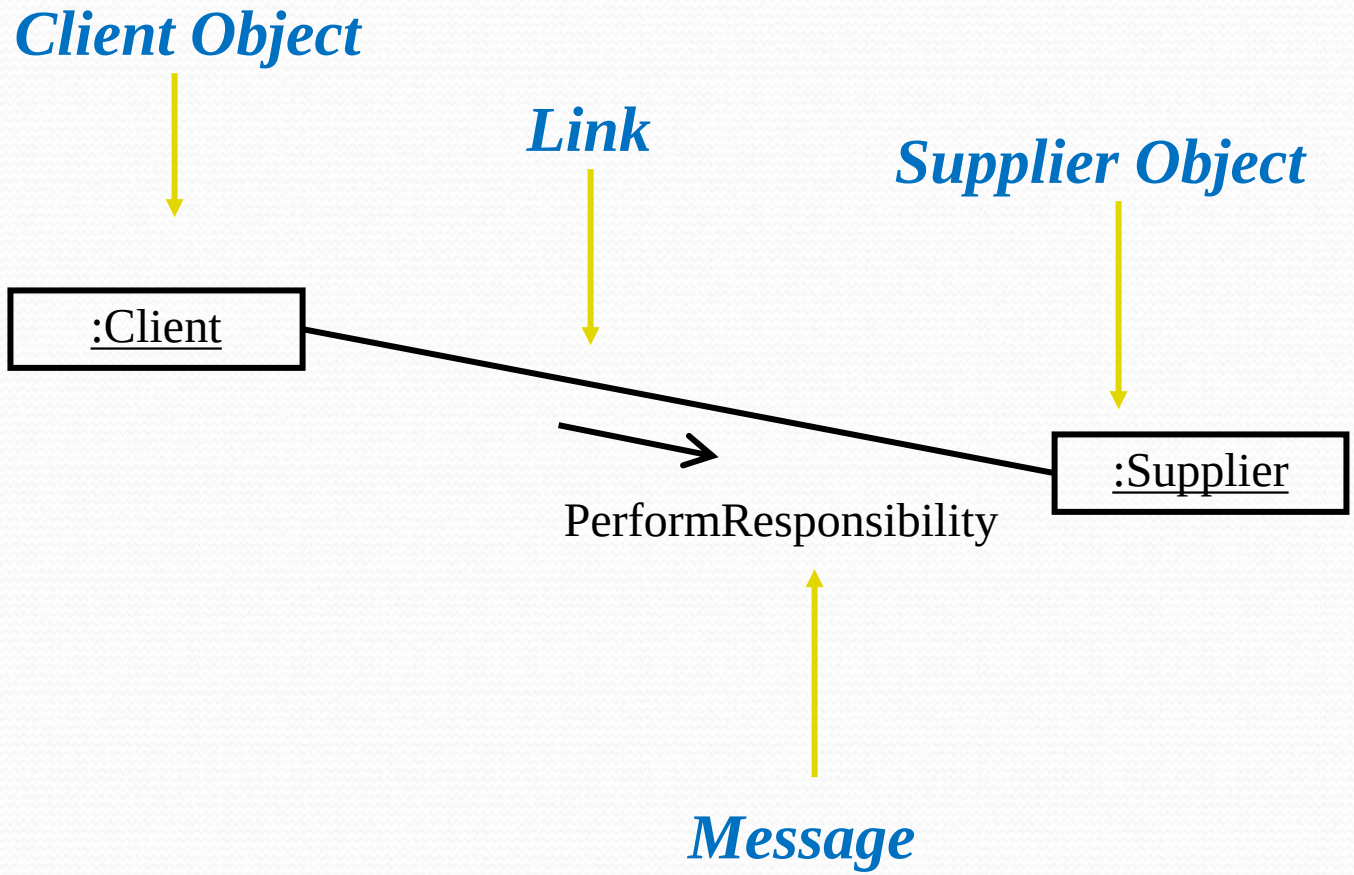
Example 1: Report Emergency




Manage EmergencyControl
IncidentForm Boundary
Incident
Acknowledgment
Dispatcher
submitReportToDispatcher()
create()
view()
createIncident()
create()
submit()
create()


FieldOfficer
Manage EmergencyContro
ReportEmergency Control
Acknowledgment Notice
acknowledgeReport()
create
view

The Anatomy of Collaboration Diagrams



Collaboration Diagrams vs. Sequence Diagrams

- Collaboration Diagrams
  - Show relationships in addition to interactions
  - Better for visualizing patterns of collaboration
  - Better for visualizing all of the effects on a given object
  - Easier to use for brainstorming sessions
- Sequence Diagrams
  - Show the explicit sequence of messages
  - Better for visualizing overall flow
  - Better for real-time specifications and for complex scenarios

Recap

- Sequence diagram is used to realize the behavior of a use case as a set of interacting objects
- The first step to identify different types of objects
- Conceptual classes are identified at this stage